Future Scope & Upgradation Plan

(For Blockchain-Based Land Registry System – 8th Semester Major Project)

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1. Purpose of This Document

- To outline how the current academic prototype can be upgraded into a **real-world, product-ready system**
- To identify **feasible technical enhancements** and **future scope**

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2. Current System Overview

- Blockchain-based land registry prototype
- Core functionalities:
  - Land record creation
  - Ownership verification
  - Ownership transfer using blockchain
- Uses smart contracts to ensure immutability and transparency
- Designed primarily as a **proof-of-concept**

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3. Limitations of Current System

- No real-world authority hierarchy
- Limited user role differentiation
- No document storage strategy for large files
- Smart contracts are static and non-upgradeable
- No performance or cost evaluation
- Deployment limited to development environment

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4. Proposed Upgradations & Future Scope

4.1 Role-Based Authority Workflow

- **Current limitation:** Single or basic admin-level control
- **Upgradation:**
  - Define multiple roles:
    - Citizen / Landowner
    - Registrar
    - Government Authority
    - Auditor
  - Multi-step approval workflow for land transfer
- **Real-world relevance:**
  - Reflects actual land registration process
  - Prevents unauthorized ownership changes

4.2 Off-Chain Document Storage with On-Chain Verification

- **Current limitation:** No efficient handling of land documents
- **Upgradation:**
  - Store documents off-chain (IPFS / cloud storage)
  - Store cryptographic hash and metadata on blockchain
  - Verify document integrity using hash comparison
- **Real-world relevance:**
  - Reduces blockchain storage cost
  - Supports large document handling

4.3 Smart Contract Audit Logging & Upgrade Awareness

- **Current limitation:** Static contracts with limited traceability
- **Upgradation:**
  - Add event-based logging for all critical actions
  - Maintain transaction history for auditing
  - Design concept for upgradeable smart contracts
- **Real-world relevance:**
  - Legal traceability
  - Supports long-term system evolution

4.4 Hybrid Blockchain Architecture (Conceptual)

- **Current limitation:** Single public blockchain usage
- **Upgradation:**
  - Public blockchain:
    - Ownership proof
    - Verification hashes
  - Permissioned blockchain:
    - Transaction details
    - Authority-level access
- **Real-world relevance:**
  - Data privacy
  - Regulatory compliance

4.5 Secure Identity & Authentication Layer

- **Current limitation:** Wallet-based or basic authentication
- **Upgradation:**
  - Multi-factor authentication model
  - Identity verification abstraction (simulated)
- **Real-world relevance:**
  - Prevents impersonation
  - Enhances trust in system

4.6 Performance, Scalability & Cost Analysis

- **Current limitation:** No system evaluation metrics
- **Upgradation:**
  - Measure:
    - Transaction latency
    - Gas cost per operation
  - Analyze scalability under increased usage
- **Real-world relevance:**
  - Determines economic feasibility
  - Important for government-scale deployment

4.7 Deployment & Infrastructure Readiness

- **Current limitation:** Local development environment
- **Upgradation:**
  - Cloud-based deployment
  - Environment separation (development / testing)
  - Basic infrastructure planning
- **Real-world relevance:**
  - Enables real-user access
  - Improves system reliability

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5. Phase-Wise Implementation Plan

- **Phase 1:**
  - Role-based workflow
  - Smart contract enhancement
- **Phase 2:**
  - Document storage integration
  - Audit logging
- **Phase 3:**
  - Performance analysis
  - Deployment testing

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6. Expected Outcomes

- Product-ready land registry architecture
- Improved security, transparency, and traceability
- Scalable and cost-aware blockchain solution
- Alignment with real-world land administration systems